CBE 179 Cheat Sheet

Ashutosh Bhadauria,

+ CZ growth

$$\frac{\partial^2 \tilde{T}_s}{\partial \tilde{y}^2} = Pe \frac{\partial \tilde{T}_s}{\partial \tilde{y}} - N_R(\tilde{T}_s^4 - \tilde{T}_\infty^4) - N_B(\tilde{T}_s - \tilde{T}_\infty) = 0 \quad ; \quad \tilde{T}_s = \frac{T_s}{T_0} \; ; \; \tilde{y} = \frac{y}{R} \; , \; \tilde{T}_\infty = \frac{T_\infty}{T_0} \; ; \; N_R = \frac{2\sigma \varepsilon_0 R T_0^3}{k_s} \; ; \; N_B = \frac{2hR}{k_s}$$

Assumptions: → U_s is small ($Pe \sim 0$); → Axial conduction ($N_R \gg N_B$) ≫ Convective transport; → $\tilde{T}_s \gg \tilde{T}_\infty$

$$\left.\begin{aligned}\frac{\partial^2 \tilde{T}_s}{\partial \tilde{y}^2} &= N_R \tilde{T}_s^4 \\ \Rightarrow \frac{u_s^{max} R \rho_L c_{p_L}}{k_L} &= Pe_L^{max} = \frac{2}{\sqrt{10}} \tilde{T}_m^{5/2} S^{-1} \tilde{K} (N_B)^{1/2} \frac{\rho_L}{\rho_S}\end{aligned}\right\} \quad \tilde{K} = \frac{k_s}{k_L} \; ; \; S = \Delta H_s / c_{p_L} T_0$$

Production Rate: $P = \mu_s A \rho_s$; $P \propto R^{3/2}$; $C_{A,S} = k_0 C_{A0} (1-x)^{k_0 - 1}$; $x \rightarrow$ fraction of melt in solid

+ Doping

$D = D_0 \exp\left(\frac{-E_a}{k_B T}\right)$

① Pre-deposition (constant C_s)

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} \Rightarrow C(x,t) = C_s \operatorname{erfc}\left[\frac{x}{2\sqrt{Dt}}\right] \quad ; \quad \operatorname{erfc} = 1 - \frac{2}{\sqrt{\pi}} \int_0^y e^{-y^2} dy$$

BC: 1) $C(x,t) = 0$ 2) $C(\infty, t) = 0$ 3) $C(0,t) = C_s$ (IC)

$$Q(t) = \int_0^\infty C(x,t)\, dx = \frac{2}{\sqrt{\pi}} C_s \sqrt{Dt} \quad ; \quad x = 2\sqrt{Dt} \operatorname{erfc}^{-1}\left(\frac{C(x,t)}{C_s}\right)$$

② Drive in

$$C(x,t) = \frac{S}{\sqrt{\pi D t}} \exp\left(\frac{-x^2}{4Dt}\right) \quad ; \quad \text{BC: 1) } C(\infty, t) = 0 \; ; \; 2) \int_0^\infty C(x,t)\, dx = S = \text{Total amount of dopants/area}$$

Characterisation

- 4-point probe: $R = \left(\frac{\rho}{t}\right)\left(\frac{L}{W}\right)$; t = thickness; $R_s = \frac{1}{q \int \mu \cdot C(z)\, dz} = k\left(\frac{V}{I}\right)$; (If probe space ≫ Relevant thickness of film)
- SIMS (Secondary Ion Mass Spect.) → counts per "sputter" time gives you depth → defects identity of sputtered ions via Mass Spec
  - Pros: • Super precise • Detect everything
  - Cons: • Hard to calibrate • Damages layer



$$-D_{AB} \frac{d^2 C_A}{dz^2} = 0 \quad ; \quad C_A(z) = C_1 z + C_2 \quad ; \quad C_A(z) = \frac{C_S - C_S}{S} z + C_{A,S}$$

BC: 1) $C_A(z=0) = C_S$ 2) $C_A(z=S) = C_S(t)$

$$-D_{AB} \frac{dC_A}{dz} = k C_S(t) \quad \left[\text{Flux} = \frac{\text{Rate of } O_2 \text{ consumed}}{}\right] \quad \left\{ C_S(t) = \frac{D_{AB} C_S}{(kS + D)} \right.$$

$$R|_S \, MW_{SiO_2} = \rho_{SiO_2} \frac{dz}{dt} \quad \Rightarrow \quad \boxed{\frac{dz}{dt} = \frac{k D C_S \bar{V}}{kS + D}} \quad \Rightarrow t = \frac{z}{k_{lin}} + \frac{z^2}{k_{par}} \; ; \; k_{par} = (2 D_{AB} C_S \bar{V}) \; ; \; k_{lin} = (k C_S \bar{V})$$

$$\left(\frac{dz}{dt}\right)^{-1} = \frac{1}{k_{lin}} + \frac{2}{k_{par}} t \; ; \; k = A \exp\left(\frac{-E_a}{k_B T}\right)$$

+ Thermal Oxidation

$\bar{V} = \frac{MW}{\rho}$; $\frac{\text{d of } Si \text{ consumed}}{\text{d of } SiO_2 \text{ produced}} = \frac{\bar{V}_{Si}}{\bar{V}_{SiO_2}}$

$k_{lin} \propto$ crystallographic orientation

+ Kinetic Theory of gases

$$f(v_x, v_y, v_z) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left(\frac{-m(v_x^2 + v_y^2 + v_z^2)}{2 k_B T}\right)$$

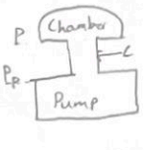
$$\bar{V} = \sqrt{\frac{8 k_B T}{\pi m}} \; ; \; \bar{E} = \frac{3}{2} k_B T \qquad f(v) = 4\pi v^2 \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left(\frac{-m v^2}{2 k_B T}\right)$$

$$P = n k_B T \; ; \; \lambda^{mfp} = \frac{1}{n\sigma} \; ; \; \tau = \frac{1}{n\sigma\bar{v}} \qquad f(E) = 2\left(\frac{E}{\pi}\right)^{1/2} \frac{1}{(k_B T)^{3/2}} \exp\left(\frac{-E}{k_B T}\right)$$

$\nu = n\sigma\bar{v}^{-1}$; n = #molecules/Volume; σ = cross sect. area; τ = mean collision time

$k_B = \frac{R}{N_A}$

$J_e \rightarrow$ Evaporation flux [=] molecules/cm²·s → [=] g/cm²·s



$S = Q/P$; $C = 12.2 \frac{D^3}{L}$; $S = \frac{S_p}{1 + \frac{S_p}{C}}$

+ Evaporation

$$\Gamma_e = 5.83 \times 10^{-2} \sqrt{\frac{M}{T}} \times (P_e)[=] \frac{g}{cm^2 \cdot s}$$

$$J_e = \frac{P_e}{\sqrt{2\pi m k_B T}} \; ; \; \Phi_e = \frac{N_A P_e}{\sqrt{2\pi M R T}} \; (M \text{ in kg/mol})$$

$$\frac{\Phi_A}{\Phi_B} = \frac{\gamma_A x_A P_A^{Sat}}{\gamma_B x_B P_B^{Sat}} \sqrt{\frac{M_B}{M_A}} = \frac{x_A / MW_A}{(1 - x_A)/MW_B}$$

(Composition of the alloy evaporated; $\rightarrow$ composition of alloy source (target))

↳ Point Source

$$\bar{M}_e = \int_{t_{nc}}^{t} \Gamma_e \, dA_e \, dt \; ; \; \frac{d\bar{M}_s}{dA_s} = \frac{\bar{M}_e}{4\pi} \frac{\cos\theta}{r^2} \; ; \; d(l) = \frac{\bar{M}_e h}{4\pi (h^2 + l^2)^{3/2}} \Rightarrow \frac{d(l)}{d_0} = \frac{1}{\left[1 + \left(\frac{l}{h}\right)^2\right]^{3/2}}$$

↳ Surface Source

$$\frac{d\bar{M}_s}{dA_s} = \frac{\bar{M}_e \cos\theta \cos\phi}{\pi r^2} \Rightarrow \frac{d\bar{M}_s}{dA_s} = \frac{\bar{M}_e}{4\pi r_0^2} \Rightarrow \frac{d(l)}{d_0} = \frac{1}{\left[1 + (l/h)^2\right]^2}$$

+ Sputtering

$$S = \frac{(\#\text{ of ejected atoms})}{\text{incident ion}} = \frac{\bar{E}}{E_t}\left(1 + \left[\frac{\ln\left(\bar{E}/E_b\right)}{\ln 2}\right]^{1/2}\right) \sigma_0 \cdot n_{areal}$$

(E_b = surface binding energy; σ_0 = collision cross section; n_{areal} = SA density)

$\sigma_0 = \pi a^2$; $a = \frac{a_b}{\sqrt{Z_1^{2/3} + Z_2^{2/3}}}$

$\bar{E} = \frac{F_2 + E_1}{2} E_1$ (E_1 = Energy of incident ion; $\bar{E}$ = KE transferred to target atom in primary collision)

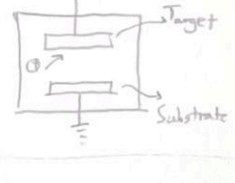
$E_2 = \frac{4 M_1 M_2}{(M_1 + M_2)^2}$

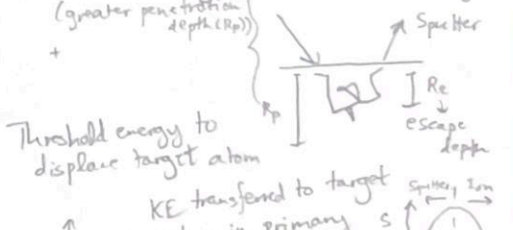
Alloys

$\frac{C_A^f}{C_B^f} = \frac{C_A S_B}{C_B S_A}$; $\frac{C_A}{C_B} = \frac{x_A / MW_A}{x_B / MW_B}$ (wt%)

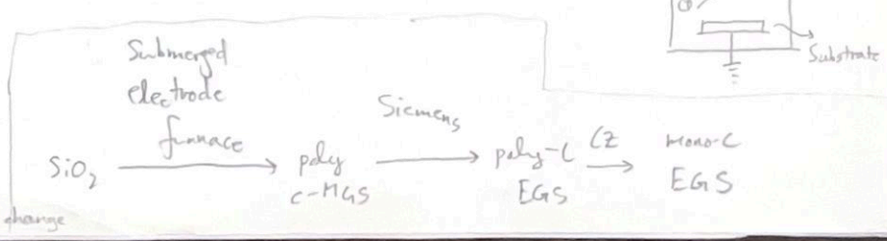
C_A^f / C_B^f: target mole fraction. Film on substrate has the same composition = Bulk target composition

C_A / C_B: Alloy mole fraction. However surface target comp will change





Threshold energy to displace target atom



PVD

+ No chemical rxn
+ Uses solid sources

$J_x = n\bar{v}_x = \frac{1}{4} n \bar{V}$; $n = \frac{P}{k_B T}$

↳ Vaccum $J_z = \frac{P}{\sqrt{2\pi m k_B T}}$; $\tau = \frac{(n_{areal})}{J_z \cdot S}$, $n_{areal} = \frac{(\#\text{ Atoms in plane})}{a^2}$

S → sticking factor

$a = \left[\frac{\bar{V}}{N_A}(\#\text{ Atoms in lattice})\right]^{1/3}$

D_{Diff} coefficient

$D_A = \frac{1}{3} \bar{v} \lambda = \frac{1}{3} \sqrt{\frac{8 k_B T}{\pi m}} \frac{kT}{P} \frac{1}{\sigma}$; $Q = C(P_2 - P_1) = P\dot{V}$

Knudsen Diff.

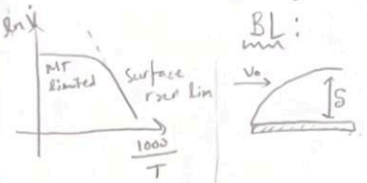
+CVD

Film growth rate → $\dot{V} = \frac{k_s h_g C_T}{k_s + h_g N_1} y$; y = mole fraction

$k_s \rightarrow$ rxn const

$h_g \rightarrow$ MT coefficient

$N_1 \rightarrow$ Bulk density

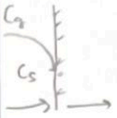


BL:

$\bar{\delta} = \frac{10}{3}\frac{L}{\sqrt{Re_L}}$; $Re_L = \frac{\rho v_0 L}{\mu}$

$h_g = \frac{D_g}{\bar{\delta}} = \frac{3}{10} D_g \sqrt{\frac{\rho v_0}{\mu L}}$

$h_g \sim \sqrt{q}$ → volumetric flow rate



Diffusion ∝ $\frac{1}{\text{Pressure}}$

LPCVD: $\uparrow h_g$, $P\downarrow$, $Re\downarrow$, $\rho\downarrow$, $V_0\uparrow$, $\bar{\delta}\uparrow$, $D\uparrow$, $\mu \sim$ const

More uniform growth rate as $\dot{V}$ is reaction limiting

MOCVD: Used to deposit thin film of compound semiconductors

Thermo

$P^\circ = 1$ bar (ref)

$G(T,P,n_j) = \sum_{i=1}^{N} n_i \mu_i$; $\mu_i = \mu_i^\circ + RT \ln a_i$; $a_i = \left(\frac{n_i}{n}\right)\left(\frac{P}{P^\circ}\right)$

@ eq. $\Delta G = -RT \ln K$, $K = \frac{P_C^c P_D^d}{P_A^a P_B^b}$

$\frac{dG}{d\xi} = 0$; $\dot{n}_A = n_{A0} - \upsilon_A \xi$; $F = N + 2 - \pi$

CVD Cons { Typically toxic / flamable, eg AsH_3; pyrophonic reactants / products

ALD (1 layer): 1) Pulse $Al(CH_3)_3$
2) Purge (Self limiting reaction as substrate is used up)
3) Pulse H_2O
4) Purge

+ALD: Sequential use of gasphase reactions

Pros: Very controlled, precise thickness, conformality, lower deposition T

Cons: Impractical for thick films, inferior film quality, complex

horizontal & vertical surfaces are coated to the same thickness

$A = 1 - \frac{R_L}{R_V}$ ($A \sim 1$ is anisotropic which is what we want)

+Etching (Etch rate, Selectivity, Undercut)

(uniformity)

Wet

immersed in solution (HF)

Pros
+ Highly selective (higher etch rate)
+ Low cost
+ Does not damage

Cons
+ Hard to control (Start & stop)
+ Isotropic etching
+ Contamination
+ Large chemical waste



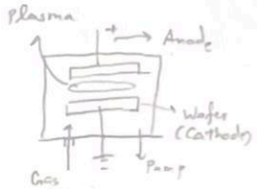
→ Direction etching possible w/ single crystalline materials (due to diff etching rate of planes)

Dry

Plasma: partially ionized gas consisting of e^-, ions, radicals & neutral molecular species. Low-P which undergo:

1) Dissociation: $e^- + AB \rightleftharpoons A + B + e^-$
2) Atomic ionization: $e^- + A \rightleftharpoons A^+ + 2e^-$
3) Molecular ionization: $e^- + AB \rightleftharpoons AB^+ + 2e^-$
4) Atomic excitation: $e^- + A \rightleftharpoons A^* + e^-$
5) Molecular excitation: $e^- + AB \rightleftharpoons AB^* + e^-$
6) Dissociative ionization: $e^- + AB \rightleftharpoons A^+ + B + 2e^-$

} these react w/ substrate to give volatile products



Cons
+ Easy to start & stop
+ Less sens. to T var.
+ Directional + Anisotropic etching

Types:
- $P \leq 10$ mTorr (Physical basis, sputtering): highly directional etching, Anisotropic, less selective (determined by sputter yield)
- $P > 100$ mTorr (chemical bases using neutral radicals): Highly selective, less damage, Isotropic
- $P \sim 10$–100 mTorr (Reative ion Etching)

$J_h = \sqrt{\frac{n^2 k_B T}{2\pi M}}$ (Flux of neutral radicals striking a surface)

Reactors:
- DC Parallel Plate, RF Discharge Plasma (for insulators), High density Plasma / Electron-Cyclotron Resonance Source)
  - (6 steps ascending effect)
  - ↳ Controls average ion-bomb energy & ion flux

Etch rate: $J_T = S \cdot J$ [=] $\frac{\text{atoms}}{m^2 \cdot s}$

[=] $\left[\frac{m}{s}\right]$ we need $\frac{\text{atoms}}{m^3}$ unit

$\therefore \frac{\rho}{MW}$ [=] $\frac{mol}{m^3}$

$\frac{\rho}{MW} N_A$ [=] $\frac{\text{atoms}}{m^3}$

$\therefore$ Etch rate $= \frac{J_T}{\left(\frac{\rho}{MW} N_A\right)}$

$\vec{v} \perp \vec{B}$; $F = q\vec{v} \times \vec{B}$

$r = \frac{m_e v}{qB}$

Cyclotron freq $\omega_c = \frac{v}{r_c} = \frac{qB}{m_e}$

$M \cdot k_B = \frac{(M)^{g}_{mol} \cdot 10^{-3}}{6.023 \times 10^{23}} \cdot 1.38 \times 10^{-23} = M \cdot (2.29 \times 10^{-50})$

CBE 179 MT1 Review

$n \Rightarrow$ Electron concentration (conc. of e^- in the conduction band)

$$\boxed{n = N_c \exp\left(\frac{-(E_c - E_F)}{kT}\right)}$$

; $N_c = 2\left[\frac{2\pi m_n kT}{h^2}\right]^{3/2}$ ← effective density of states of the conduction band.

$$\boxed{p = N_v \exp\left(-\frac{(E_F - E_v)}{kT}\right)}$$

$N_v = 2\left[\frac{2\pi m_p kT}{h^2}\right]^{3/2}$ (valence band)

* Probability of finding an e^- at a certain energy:

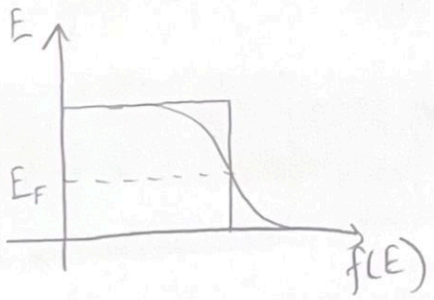
$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

$$f(E) \approx \exp\left(\frac{-(E - E_F)}{kT}\right)$$ (Boltzmann approximation)

→ for large E: $E - E_F \gg kT$

→ for small E: $E_F - E \ll -kT$

$$f(E) \approx 1 - \exp\left(\frac{-(E_F - E)}{kT}\right)$$



All equations are derived assuming this assumption $(E - E_F) > 3k_BT$ i.e. does not work for very heavily doped semiconductor

* np & Intrinsic Carrier Conct (n_i)

$$np = N_c N_v e^{-(E_c - E_v)/kT}$$

$$\boxed{np = n_i^2}$$

The np product is constant for a semiconductor (no matter if doped or undoped)

majority vs minority Carriers:

N-type: e^- (Major), holes (Minor)

P-type: e^- (Minor), holes (major)

For intrinsic Semiconductors:

$n = p = n_i$

(electrons & holes are created equally)

Fermi level for intrinsic semiconductors:

Intrinsic semiconductor has $n_0 = p_0$

$$\therefore N_c \exp\left(\frac{-(E_c - E_F)}{k_B T}\right) = N_V \exp\left(\frac{-(E_F - E_V)}{k_B T}\right)$$

$$\ln N_c + \frac{-(E_c - E_F)}{k_B T} = \ln N_V + \frac{-(E_F - E_V)}{k_B T}$$

$$\frac{E_F - E_c + E_F - E_V}{k_B T} = \ln N_V - \ln N_c$$

$$\frac{2E_F}{k_B T} - \frac{(E_c + E_V)}{k_B T} = \ln \frac{N_V}{N_c}$$

$$E_i = E_F = \frac{(E_c + E_V)}{2} + \frac{k_B T}{2} \ln\left(\frac{N_V}{N_c}\right)$$

$$(E_i - E_g)_m = \frac{k_B T}{2}\left(\frac{3}{2}\right) \ln\left(\frac{m_p^*}{m_n^*}\right)$$

If $m_p^* \geq m_n^*$; $E_i = E_{g_m}$ (exactly in the middle)

If $m_p^* \neq m_n$; the density of states dictates where the fermi level will be located.

As discussed before, thermal equilibrium conc. of electrons & holes is given by:

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

$$n = N_c \exp\left[\frac{-(E_c - E_i)}{kT}\right] \exp\left[\frac{E_F - E_i}{kT}\right]$$

(n_i)

$$\boxed{n_0 = n_i \exp\left[\frac{(E_F - E_i)}{kT}\right]}$$

n_i = intrinsic carrier conc.

Similarly, $\boxed{p_0 = n_i \exp\left[\frac{(E_F - E_i)}{kT}\right]}$

$$\therefore n_0 p_0 = N_c N_V \exp\left[\frac{-E_g}{kT}\right] = n_i^2$$

$$n_i = N_c \exp\left[\frac{-(E_c - E_i)}{kT}\right] = \sqrt{N_c N_V} \exp\left[\frac{-E_g}{2kT}\right]$$

(For intrinsic)

$$E_c - E_F \approx E_F - E_V$$

$$E_c + E_V \approx 2E_F$$

$$E_F \approx \frac{E_c + E_V}{2} \approx \frac{E_c}{2} - \frac{E_g}{2}$$

$$E_g = E_c - E_V$$

$$\frac{(E_c + E_V)}{2} = \frac{E_c}{2} - \frac{E_g}{2} = E_{g_m}$$

Midway between E_g

If we are to express these in terms of the intrinsic fermi level: We can add & subtract E_i

$$-(E_c - E_i) + (E_F - E_i)$$

N Junction

P & N layers are uniformly doped

Complete ionization (i.e. there is no

There are 4 charged specis in a sem

In N-type: $N_d - N_A \gg n_i$; n

P-type: $N_a - N_d \gg n_i$; p

If $N_a \gg N_d$ then it just becomes, p

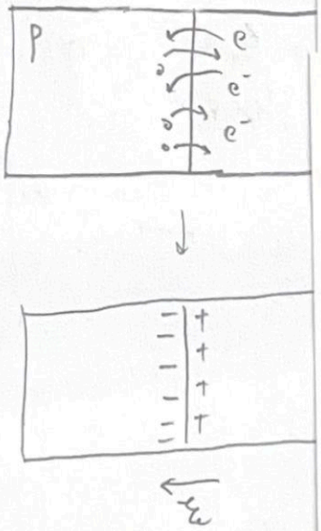
or $N_d \gg N_a$

+ We only consider abrupt junctio

two semiconductors

I_{drift} (Mino

I_{diff} (Major)



(4) Therefore, a space charge develops creating a depletion region ↓

inhibits further e⁻ transfer

[↳ e⁻ are repelled by negative charge & attracted to the positive charge

↳ Therefore an equilibrium is achieved.

@ equilibrium, there

I_{drift}^{minor} (flow of ...

w/ I_{diff}^{major} leading to net $J^{total} = 0$ [Total $I = 0$)

$$J_n^{drift} = q n \mu_n E$$

$$J_n^{diff} = q D_n \frac{dn}{dx}$$

$$J_n^{total}(x) = \underbrace{q n \mu_n E}_{drift} + \underbrace{q D_n \frac{dn}{dx}}_{diff.}$$

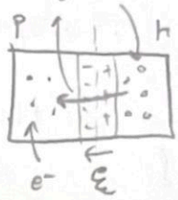
$$J_p^{total}(x) = q p \mu_p E - q D_p \frac{dp}{dx}$$

$$\boxed{\frac{D}{\mu} = \frac{kT}{q}} \rightarrow \text{Einstein relationship}$$

$$\boxed{V_b = \frac{k_B T}{q} \ln \frac{P_p}{P_n}} \approx \begin{matrix} + \text{ ionized} \\ + N_d \gg N_a \\ n = N_d \end{matrix} \quad V_b = \frac{k_B T}{q} \ln \left[(N_a N_d \right.$$

WHAT'S YOUR POINT? The barrier acts as a downhill for minority charge carriers (as they are moving to a lower voltage diff.)

I_{drift} holes



Discover the best

WHAT'S YOUR POINT? Positive Bias vs Negative Bias

+ V_b decreases
+ Move diffusion
+ Decreasing width
+ Drift more or less the same
+ High I

+ V_b increases.
+ Drift same
+ Increasing W
+ Attracts the major charge to the col
↳ No diff.
+ Low or zero I

Discover the best
Pentel

WHAT'S YOUR POINT? Zener vs. Avalance

+ highly doped (low V)
+ Charge carriers just tunnel.

+ Reverse bias (high V)
+ minority charge carriers form more e⁻-hole
↳ 'Impact Ionize'

Discover the best
Pentel.

Junction

P & N layers are uniformly doped
Complete ionization (i.e. there is no IE and the E_d/E_a is equal to E_v & E_c resp.)

There are 4 charged specis in a semiconductor: e^-, p, D^+, A^- → acceptor ions
↑ Donor atoms

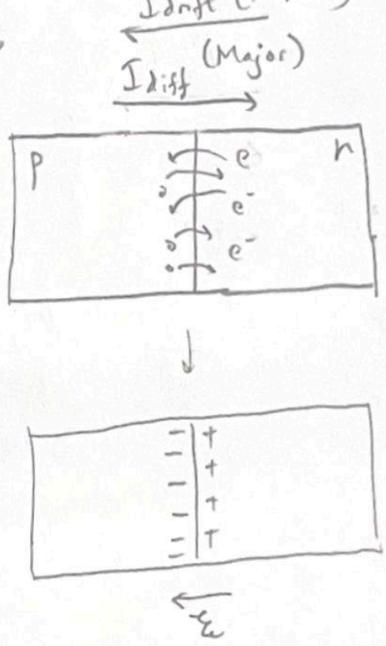
In N-type: $N_d - N_A \gg n_i$; $n = N_d - N_a$

P-type: $N_a - N_d \gg n_i$; $p = N_a - N_d$

} $n_i^2 = np$ (still holds)

If $N_a \gg N_d$ then it just becomes, $p = \frac{n_i^2}{N_d}$; $n = \frac{n_i^2}{N_a}$
or $N_d \gg N_a$

+ We only consider abrupt junction where N_a & N_d are present in two semiconductors



(1) n & p are charge nuetral
(2) When conc. gradient is established, the majority charge carriers move to the other side to recombine.
(3) This leaves the middle region depleted of carriers and therefore becomes charged. (n side is positive due to losing an e^- and p-side is negative due to losing a hole)

(4) Therefore, a space charge develops creating a depletion region ↓
inhibits further e^- transfer
[↳ e^- are repelled by negative charge & attracted to the positive charge
↳ Therefore an equilibrium is achieved.

@ equilibrium, there is still flow of charges. However, the I_{diff}^{minor} (flow of holes from n-side to p side) is balanced w/ I_{diff}^{major} leading to net $J^{total} = 0$ [Total $I = 0$)

$$J_h^{drift} = q p \mu_n E$$

$$J_n^{diff} = q D_n \frac{dn}{dx}$$

$$J_n^{total}(x) = \underbrace{q n \mu_n E}_{drift} + \underbrace{q D_n \frac{dn}{dx}}_{diff.}$$

$$J_p^{total}(x) = q p \mu_p E - q D_p \frac{dp}{dx}$$

$$\boxed{\frac{D}{\mu} = \frac{kT}{q}} \rightarrow \text{Einstein relationship}$$

$$\boxed{V_b = \frac{k_B T}{q} \ln \frac{P_p}{P_n}} \approx \begin{matrix} + \text{ ionized} \\ + N_d \gg N_a \\ n = N_d \end{matrix} \quad V_b = \frac{k_B T}{q} \ln \left[(N_a N_d$$

Bohr Shit: CphA

1) Electrons are in stable circular orbits (dictated by classical mechanics & Colombs Law)
2) e^- absorb and give out a photon to move in quantized orbits
3) Angular momentum (P_θ)

Centripetal = Coulombic force

$$\frac{m_e v^2}{r} = \frac{Z k_e e^2}{r^2}$$

$$v^2 = \frac{Z k_e e^2}{m_e r}$$

$$E = KE + PE$$

$$= \frac{1}{2} m_e v^2 - \frac{k_e e^2}{r}$$

$$= \frac{1}{2}\frac{k_e e^2}{r} - \frac{k_e e^2}{r}$$

$$E = -\frac{1}{2}\frac{k_e e^2}{r}$$

Quantized orbits:

$$m_e v r = \frac{nh}{2\pi}$$

$$v = \frac{nh}{(2\pi) m_e r}$$

$$\frac{n^2 h^2}{4\pi^2 m_e^2 r^2} = \frac{Z e^2}{4\pi\varepsilon_0 r} \Rightarrow r = \frac{\varepsilon_0 n^2 h^2}{\pi m_e e^2}$$

$$\Rightarrow \boxed{r_n = \frac{4\pi\varepsilon_0 n^2 \hbar^2}{m e q^2}} \qquad \hbar = \frac{h}{2\pi}$$

Bohr radius →

Free electron gas (FEG) model

↳ represents electronic structure of metals

Valence e^- of the crystal atoms become conduction e^- & travel freely throughout the crystal.

↳ : We neglect the interaction of conduction e^- w/ ions of the lattice & interaction between the conductive e^-. → free e-gas

↳ Charge neutrality is maintained

In 1D: n & m_s are only required to describe the electronic state
↳ each n can have upto 2 e^- (with opp spins)

The energy of the highest occupied level is called fermi energy

$$E_F = \frac{\hbar^2}{2m}\left(\frac{\pi N}{2L}\right)^2$$ [for a particle in a box]

(No e^--e^- interact is assumed)

